

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Summer Examination – 2023

Course: B. Tech.

Branch :Civil Engineering

Semester :VI

Subject Code & Name:BTCVC603

Transportation Engineering

Max Marks: 60

Date:17/07/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Explain- Stages of Engineering Survey for highways	CO1	6
B) Why the Jayakar committee was formed? Also state various recommendations of Jayakar committee.	CO1	6
C) Explain in detail- i) Camber ii) Classification of camber based on shape iii) Various values of camber for different types of road.	CO2	6
Q.2 Solve Any Two of the following.		12
A) The radius of horizontal curve is 400m, the total pavement width at curve is 7.6m (including extra widening) and the super elevation is 0.07. Check the safety against transverse skidding and calculate the transition curve length for a speed of 100kmph by using “allowable rate of introduction of super elevation” criteria ONLY . Assume pavement to be rotated about the inner edge and rate of introduction of super elevation is 1 in 150.	CO1	6
B) i) Define- Elongation Index and Flakiness Index ii) Explain the procedure of Flash and fire test of bitumen	CO2	6
C) Explain California Bearing ratio test on Subgrade	CO2	6
Q. 3 Solve Any Two of the following.		12
A) i) Write a short note on Road Arboriculture. ii) Give the difference between Tar and Bitumen.	CO3	6
B) Descending grade of 1 in 30 and ascending grade of 1 in 40 meet to form valley curve. Design a valley curve for fulfillment of comfort criteria and head light sight distance criteria if design speed is 70 kmph, $f=0.35$ and braking time is 2.5 sec, allowable rate of change of acceleration is 0.6m/sec^3	CO3	6
C) Explain road intersections at grade with neat sketches.		6

Q.4 Solve Any Two of the following.		12
A) The speed of overtaking and overtaken vehicles is 80 and 50 kmph respectively on a two way traffic road. The average acceleration during overtaking may be assumed as 0.90m/sec^2 and length of wheel base is 6m. Calculate safe overtaking sight distance for two way traffic.	CO4	6
B) Give a short note on Accident -its classification and causes	CO3	6
C) Explain i) Traffic capacity and its types ii) Urban road classification	CO3	6
Q. 5 Solve Any Two of the following		12
A) Give the difference between flexible pavements and Rigid pavements	CO3	6
B) Define Intelligent Transport System (ITS) and give some examples of Intelligent Transport System (ITS) in India.	CO4	6
C) Calculate radius of relative stiffness and radius of equivalent resisting section of a cement concrete pavement by using following data. Wheel load = 5100kg, Modulus of elasticity of concrete (E) = $3 \times 10^5 \text{ kg/cm}^2$, Pavement thickness(h) = 18cm, Poisson's ratio of concrete (μ)=0.15, Modulus of subgrade reaction(K)=6 kg/cm^3 , Radius of contact area(a)=15cm	CO4	6
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